Menu-Driven Calculator Using Python and NumPy

Problem Statement

The objective of this project is to create a calculator program that allows users to perform different mathematical operations through a menu-based interface. The program should take user input, perform calculations like addition, subtraction, multiplication, division, power, and modulus, and display results. It should also use NumPy for at least one operation and allow the user to continue until they choose to exit.

Logic Explanation

- First, display a menu with all available operations.
- Take the user's choice as input.
- Depending on the choice:
- Ask for the required numbers.
- Store them in a list.
- Perform the selected operation using functions.
- Use NumPy for multiplication.
- Handle errors like division by zero.
- Display the result.
- Ask the user whether they want to continue or exit.

CODE

```
import numpy as np

def addition(nums):
    return sum(nums)

def subtraction(nums):
    result = nums[0]
    for n in nums[1:]:
        result -= n
    return result

def multiplication(nums):
    return np.prod(nums)   # Using NumPy here

def division(nums):
    result = nums[0]
    for n in nums[1:]:
```

```
        if n == 0:
            return "Error: Division by zero!"
        result /= n
    return result

def power(base, exp):
    return base ** exp

def modulus(a, b):
    if b == 0:
        return "Error: Modulus by zero!"
    return a % b

while True:
    print("\n===== MENU DRIVEN CALCULATOR =====")
    print("1. Addition")
    print("2. Subtraction")
    print("3. Multiplication (NumPy used)")
    print("4. Division")
    print("5. Power")
    print("6. Modulus")
    print("7. Exit")

    choice = input("Enter your choice: ")

    if choice in ['1', '2', '3', '4']:
        count = int(input("How many numbers? "))
        nums = []

        for i in range(count):
            num = float(input(f"Enter number {i+1}: "))
            nums.append(num)

        if choice == '1':
            print("Result:", addition(nums))

        elif choice == '2':
            print("Result:", subtraction(nums))

        elif choice == '3':
            print("Result:", multiplication(nums))

        elif choice == '4':
            print("Result:", division(nums))
```

```
    elif choice == '5':
        base = float(input("Enter base: "))
        exp = float(input("Enter exponent: "))
        print("Result:", power(base, exp))

    elif choice == '6':
        a = float(input("Enter first number: "))
        b = float(input("Enter second number: "))
        print("Result:", modulus(a, b))

    elif choice == '7':
        print("Exiting calculator. Goodbye!")
        break

    else:
        print("Invalid choice! Please try again.")

    cont = input("Do you want to continue? (y/n): ")
    if cont.lower() != 'y':
        print("Exiting calculator. Goodbye!")
        break
```

Sample Output

```
 MENU DRIVEN CALCULATOR
1. Addition
2. Subtraction
3. Multiplication (NumPy)
4. Division
5. Power
6. Modulus
7. Exit

Enter your choice: 1
How many numbers? 3
```